



End-to-End ETL Data Engineering & Quantitative Mutual Fund Analytics

Bluestock Capstone Project — Final Presentation

CAPSTONE ANALYTICS TEAM

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Problem Statement & Project Objectives



The Industry Challenge

Fragmented asset logs, unscheduled transaction delays, and siloed performance metrics obscure systemic portfolio downside risks — leaving leadership without a unified view of exposure.

Core Objective

Build an automated relational ETL pipeline to unify dirty CSV raw files into a secure database layer (`bluestock_mf.db`), eliminating data gaps through rigorous normalization.

Analytics Mandate

Deploy VaR, CVaR, HHI, and Sharpe trend models to generate immediate, actionable transparency for executive leadership.

Data Architecture & Processing Scale

46K+

Daily NAV Records

Historical pricing timelines serving as the baseline for volatility engines.

32,778

Investor Transaction Logs

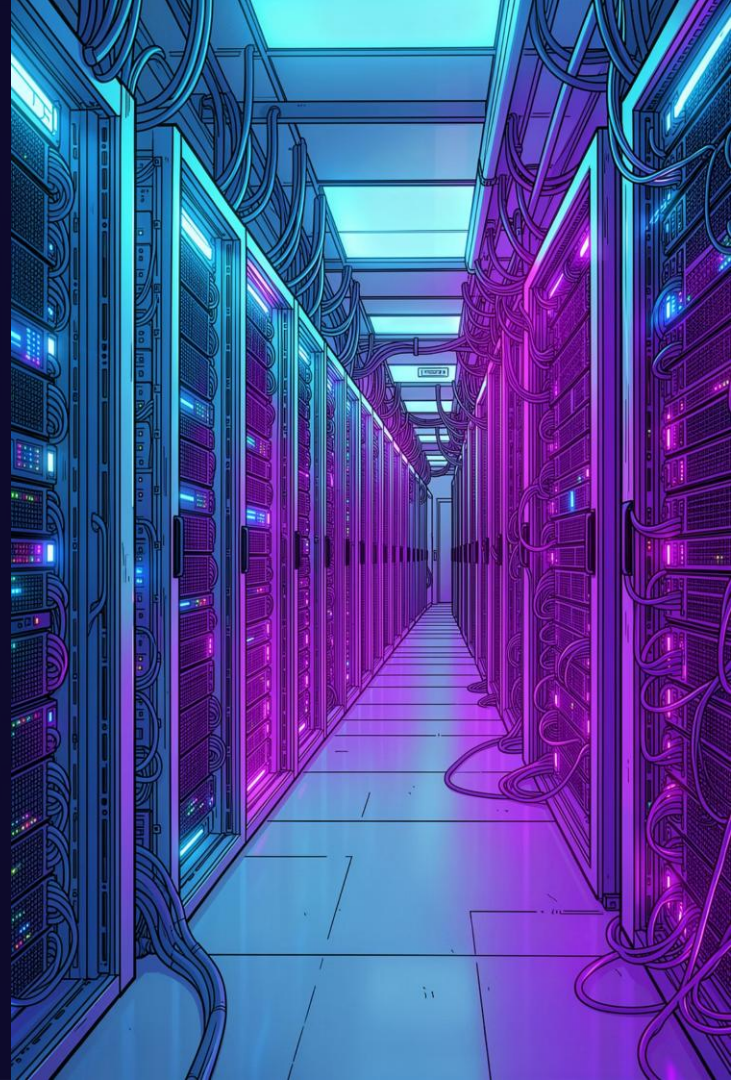
Individual client action paths tracking subscription types and volumes.

40

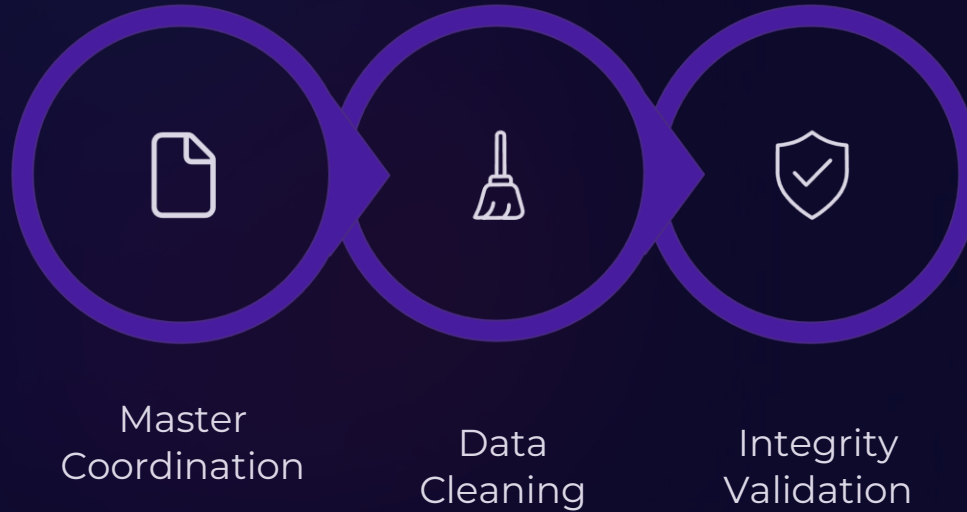
Scheme Registries

Advanced attributes across 10 asset management companies.

 **Relational Normalization:** Fields mapped using explicit primary and foreign keys, eliminating data gaps and ensuring referential integrity across all three ingested systems.

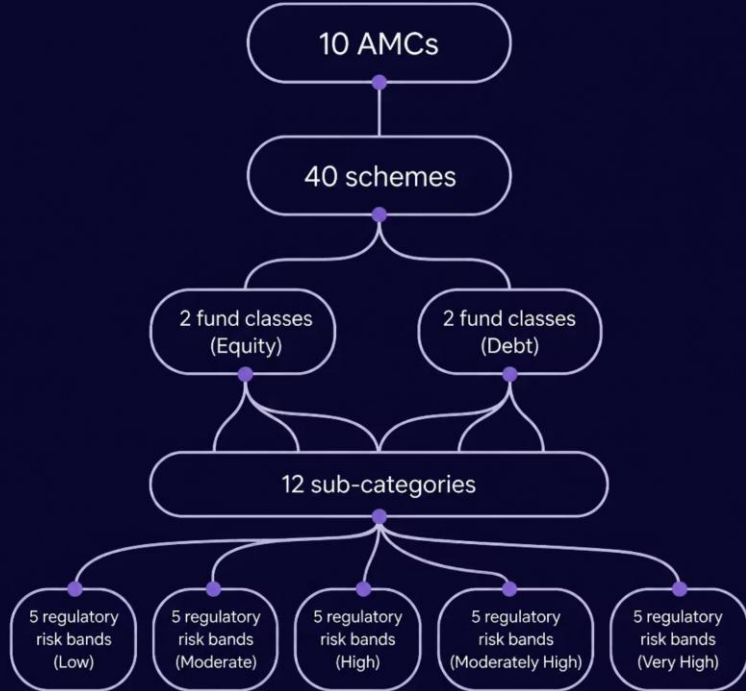


Pipeline Automation & System Integrity



The pipeline executes as a repeatable, end-to-end data lifecycle — from raw ingestion through cleaning to validated database output — with strict referential integrity safeguards at every stage.

Exploratory Data Analysis: Market Coverage



Market Asset Distribution

Monitored 40 distinct schemes across 10 top-tier AMCs, providing comprehensive coverage of the Indian mutual fund landscape.

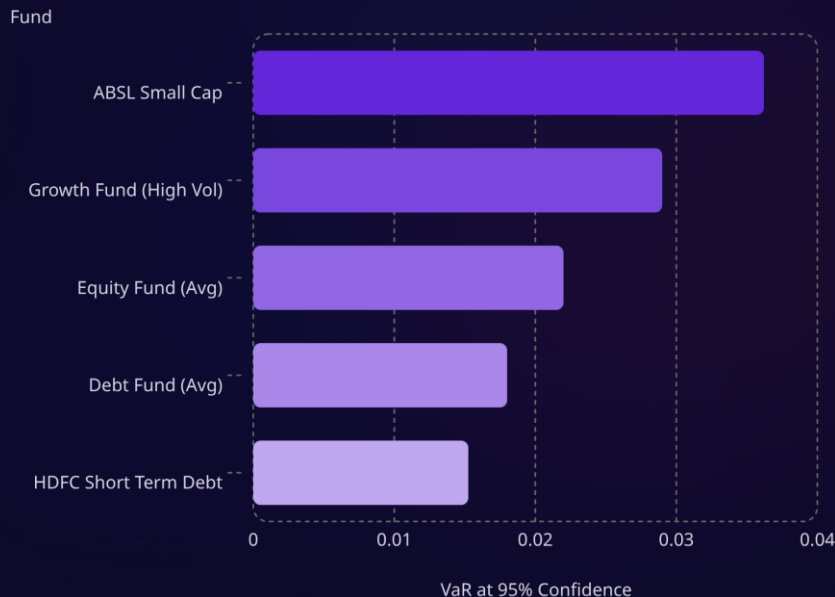
Multi-Tier Categorization

Funds structured across Equity and Debt classes, further segmented into 12 sub-categories for granular comparative analysis.

Risk Profile Segmentation

Products classified across 5 regulatory risk bands — from Low to Very High — enabling precise customer targeting and risk-aligned recommendations.

Quantitative Risk: Downside Boundaries (VaR & CVaR)



Downside Boundary Modeling

Calculated 95% Historical VaR and Expected Shortfall (CVaR) across all schemes to quantify maximum expected loss under normal market conditions.

High-Volatility Exposure

Growth and Small-Cap funds exhibit pronounced tail-risk — **ABSL Small Cap VaR at 0.036214** reflects aggressive equity holdings and elevated downside sensitivity.

Defensive Capital Havens

Fixed-income debt profiles maintain tightly bounded loss thresholds — **HDFC Short Term Debt VaR at 0.015250** demonstrates superior capital preservation.

Volatility Performance: Rolling Sharpe Trend

Annualized 90-day rolling Sharpe ratios map return-to-volatility generation over time, clearly highlighting cyclical market shifts and fund stability patterns. This visualization enables leadership to identify funds that consistently outperform on a risk-adjusted basis and those experiencing performance degradation during market stress events.



Portfolio Diversification: Sector HHI Analysis

The Concentration Framework

Implemented the Herfindahl-Hirschman Index ($\sum (\text{weight}_i)^2$) across underlying industry sectors to flag structural asset vulnerabilities before they materialize as losses.

High Concentration Warnings

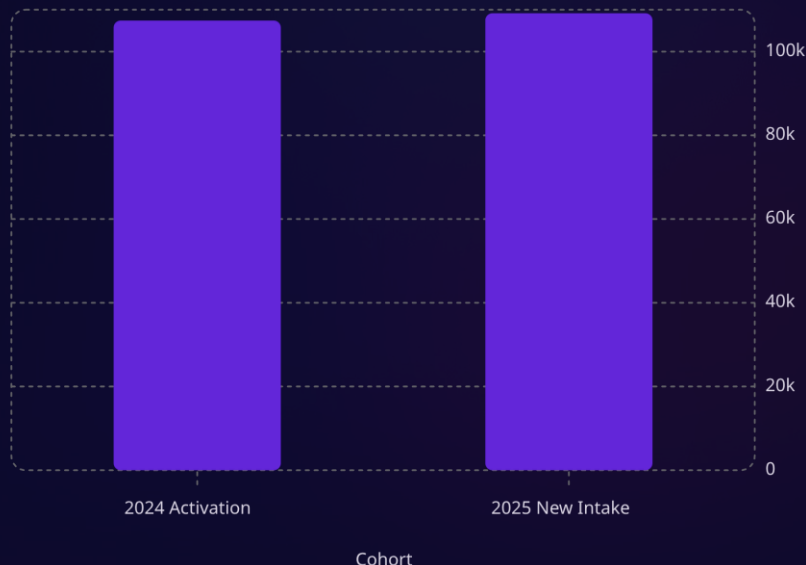
Identifies schemes carrying dangerous sector concentration, leaving them exposed to industry-specific downturns and correlated drawdowns.

Safe Diversification Vectors

Maps optimal low-HHI fund alternatives to balance risk exposure across broader industry segments, reducing systemic vulnerability.



Investor Behavior: Lifecycle & Churn Analytics



Cohort Capital Trends

Average SIP ticket sizes grew from **\$107,422** (2024) to **\$109,158** (2025), reflecting growing platform confidence and expanding capital inflows.

Critical Churn Exposure

Of 5,000 active automated accounts monitored against the 35-day continuity threshold, **4,527 breached the safety window** – a **90.54% portfolio churn risk rate** across systematic investment channels.



Individual account consistency requires proactive system alerts to protect recurring platform velocity.



Strategic Recommendations & Project Sign-Off

1

Automated Churn Mitigation

Deploy push notifications on Day 28 to alert investors before breaching the 35-day payment safety window.

2

Personalized Client Portals

Integrate `recommender.py` into the client dashboard for top-3 fund matches optimized by risk tolerance and Sharpe ratios.

3

Structural Asset Rebalancing

Rebalance portfolios away from high-concentration HHI equity layers to reduce exposure to sector-specific downturns.



Production Release: Pipeline codebase polished, documented, and officially tagged as stable version `v1.0` on GitHub. Thank you — open for questions.